Xor Circle

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	512 megabytes

CCPC (China Collegiate Programming Contest) will be held in city A. City A can be considered as an infinity 2-dimensional plane. n contestants will take part in the competition. These n people are numbered from 1 to n, each of them has an initial position in city A and a value a_i.

The organizer needs to choose a place for the competition site. For each person i, he/she will be happy if and only if the distance from the competition site to his/her initial position is not greater than d. Note that the coordinate of the site can be any real number.

To determine whether the competition site is good or not, the organizer defines $f(P)$ as the maximum xor sum of **some** of the happy people's value if point P is considered as the competition site.

Let $S(P)$ denotes all person such that distance between P and his/her location is not greater than d, $g(T)$ denotes the exclusive OR of all a_i such that person i is in T (**Specially**, $g(\varnothing) = 0$), $f(P) = \max_{T \subseteq S(P)} g(T)$. A point P is considered good if and only if $f(P) \geq k$.

You need to find out the minimum non-negative real number d, such that there are at least one location is good.

Note that the distance between point $A(x_A, y_A)$ and $B(x_B, y_B)$ is $\sqrt{(x_A - x_B)^2 + (y_A - y_B)^2}$.

Input

The first line contains two integers $n(1 \leq n \leq 1000), k(0 \leq k < 2^{20})$ - the number of the competitors and the minimum value of a good location.

For the next n lines, each line contains three integers $x_i, y_i, a_i(x_i, y_i \leq |10^4|, 0 \leq a_i < 2^{20})$ - the initial position and the value of the i-th person.

Output

A single non-negative real number d - the minimum distance such that there at least one location's value is not smaller than k.

Your answer will be considered correct if and only if the absolute or relative error is not greater than 10^{-6}.

If the answer does not exist, please output `-1`.

Examples

standard input	standard output
4 39 -1 -1 5 2 3 37 -1 2 2 -1 -1 14	1.5811388301
4 16 1 1 4 5 1 4 1 9 1 9 8 10	-1
2 0 11 45 14 191 98 10	0.0000000000